

CS 217: Practice questions on Generative Classifiers

1. Write the SVM **dual** objective for the separable case where we do not have any slack variables, that is, for this objective:

$$\begin{aligned} \min_{\mathbf{w}, w_0} \quad & \frac{1}{2} \|\mathbf{w}\|^2 \\ \text{s.t.} \quad & y_i(\mathbf{w} \cdot \mathbf{x}_i + w_0) \geq 1 \quad \forall i : 1 \dots N \end{aligned} \tag{1}$$

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2. Consider a trivially separable dataset on $d = 3$ dimensions consisting of five instances and their labels as: $D = \{([1, 2, 1], 1), ([1, 1, 1], 1), ([0, 0, 0], -1), ([-3, -1, -2], -1), ([-2, -1, -1], -1)\}$.

- (a) If you use the linear SVM max-margin formulation for the above separable case (i.e., without the slack variables ξ_i), what are the support vectors for this dataset?

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- (b) For the above case, what is the optimal value of parameters $\mathbf{w} = [w_1, w_2, w_3]$ and b ?

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- (c) If you now change the objective to minimize the L1 norm ($|w_1| + |w_2| + |w_3|$) instead of L2 norm ($w_1^2 + w_2^2 + w_3^2$), describe all possible optimal solutions for w_1, w_2, w_3 and b

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(d) Draw a compact decision tree on the above training data.

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(e) Suppose on this dataset you trained various classifiers optimally on D and now we add one more instance $(\mathbf{x}^6, y^6) = ([3, 3, 3], 1)$. For which of the classifiers below will the parameters $\mathbf{w}, b$ change after retraining with these six instances. Give a brief justification for each.

i. SVM for the separable case above.

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ii. Naive Bayes classifier

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3. If you find that SVM with a RBF kernel is over-fitting your data. State with a brief reason which of the following actions you will take:

(a) Increase or decrease C ?

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(b) Increase or decrease σ of the RBF kernel $e^{-\left(\frac{\mathbf{x}^i - \mathbf{x}^j}{\sigma}\right)^2}$?

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4. Consider a trivially separable dataset on $d = 2$ dimensions consisting of four instances and their labels as: $D = \{([1,1],1),([-1,-1],-1),([1,2],1), ([-3, -1], -1)\}$. Suppose we train a SVM on this dataset with a linear kernel and large C .

- (a) Show geometrically the training points in 2D space along with two separators — one which maximizes the separation between points of the two classes, and a second which does not.

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- (b) What is the optimal value of parameters $\mathbf{w} = [w_1, w_2]$ and b assuming C is large?

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- (c) What are the support vectors for this dataset?

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- (d) What is the value of α_i for these four points? [Hint: Do you try to solve exhaustively. Use the answer to the above question to simplify, but show the steps you followed instead of just the final answer.]

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- (e) If you now change the training objective to minimize L1 norm of $\mathbf{w}$ instead of L2 norm of $\mathbf{w}$. Note, for $d = 2$, L1 norm will be $|w_1| + |w_2|$ and L2 norm will be $w_1^2 + w_2^2$, describe all possible optimal solutions for w_1, w_2 and b

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- (f) If you now change the Hinge loss by a perceptron loss defined as follows:

$$\min_{\mathbf{w}, b} C \sum_i \max(0, -y_i(\mathbf{w}\mathbf{x}^i + b)) + ||\mathbf{w}||^2$$

what is the optimal value of the parameters $\mathbf{w} = (w_1, w_2), b$ (Assume $C > 0$)?

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- (g) If you replace the perceptron loss above by the logistic loss, does the solution that you found above continue to be optimal? State a brief reason for your answer? [Assume $C = 1$]

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5. Exercise 17.1 in Book: “Probabilistic Machine Learning” by Kevin Murphy.